

naija-asr-benchmark at Milestone 1

The harness is in better shape than the evidence it produces. Every quality gate passes and the scoring module is methodologically careful — but the control arm has never been run, and no published figure can currently be reproduced by a reader.

REPOSITORY	REVIEWED AT	DATE
lintdeveloper/naija-asr-benchmark	6c4d336 · main	24 September 2026
SCOPE		
1,613 lines, src + tests, plan v1.5		

Summary. Milestones 0 and 1 are genuinely complete and the engineering is above the standard the plan asks for at this stage. The six findings below are not code defects — the code is clean — they are gaps between what the project *claims* and what a third party could *check*. Two of them should close before Milestone 2 starts, because Milestone 2 multiplies every run by seven acoustic conditions and will make both far more expensive to fix.

01 Verification

Run locally against the working tree, not read off a CI badge.

ruff check	ruff format	mypy	pytest	CI on main
pass	pass	pass	49 / 49	green
10 rule families	17 files	16 source files	41 test functions	latest 24 Sep

The CI design is worth singling out. It pins to Python 3.11 — the `requires-python` floor — rather than “latest”, on the correct reasoning that a newer interpreter passing says nothing about the version people will actually install. Every test is network-free by construction, with the fetch path exercised through synthetic fixtures, so a flaky CDN cannot produce a red build.

02 What the numbers actually say

Read from the result JSON, not from the README.

LANGUAGE	CLIPS	REF WORDS	WER	CER	COLLAPSES	WER EXCL.	CER EXCL.
Hausa ha_ng	20	460	436.1%	316.3%	7 (35%)	106.9%	46.6%
Igbo ig_ng	5	102	226.5%	149.7%	1 (20%)	102.5%	58.1%

The interpretation in the repository is correct and worth restating, because it is the project’s first real result: a corpus WER of 436% does not mean every word is wrong. It means **insertions dwarf the references**. The model collapsed into repetition loops on 7 of 20 Hausa clips, and those seven carry the corpus figure. Excluding them puts Hausa at 106.9% — precisely the 80–100% band the plan predicted for this milestone.

The collapse rate is the finding, not the WER. That framing is right, and the code treats it as first-class rather than as a footnote: `collapse_rate` is a property on the score object, and the excluded figures are reported *alongside* the full ones, never instead of them.

03 Findings

Ranked by what each one costs if it is still true at publication.

1 The English control arm has never been run

BLOCKING

The whole argument of the plan is comparative: Nigerian-language ASR degrades *more* than English under the same conditions. That claim needs an English number measured by the same harness. There isn’t one. Yorùbá has never been run either, despite being one of the two tone-orthography arms in contribution 1.

results/	ha-whisper-tiny-*.json	x4	
	ig-whisper-tiny-*.json	x1	
	(no yo, no en)		
data/	en_us-test.parquet	383 MB	← fetched, never used
	yo_ng-test.parquet	827 MB	← fetched, never used

Both corpora are already on disk and verified. This is a few hours of laptop time, not a blocked dependency.

RECOMMENDATION

Run `en` and `yo` at 20 clips before starting Milestone 2. The English figure is the denominator for every later claim; producing it after the degradation grid invites the suspicion that the baseline was chosen to fit.

2 No published figure can be checked by a reader

BLOCKING

The README and now `plan.pdf` both state WER 436.1% and a 35% collapse rate. `results/` is in `.gitignore`, so the JSON those numbers come from is not in the repository. A reader who wants to verify them has to re-download 734 MB and re-run inference — and would get a different number if anything in the stack moved.

```
.gitignore:5  results/

ha-whisper-tiny-20260907T145421+0000.json  18 KB
```

For a project whose stated deliverable is a benchmark “reproducible by a stranger”, the evidence for its headline claim being untracked is the sharpest inconsistency in the repository. The files are 18 KB.

RECOMMENDATION

Commit the result JSONs, or attach them to a tagged release. Keep `results/` ignored for scratch runs, but a figure that appears in the README should have its artefact in the tree.

3 Runs don't record the stack that produced them

BEFORE PUBLICATION

`persist()` writes `harness_version`, the model name, the normalisation policy and the long-form strategy — thoughtful choices, all of them. It does not write the `transformers` version, the `torch` version, or the model revision SHA.

An ASR figure is a property of a model *and* the inference stack decoding it. `whisper-tiny` is a moving pointer on the Hub, and `transformers` is pinned only as `>=4.40,<5`. The repetition-collapse behaviour that is now the headline finding is exactly the kind of thing a decoding change alters. Without those three fields, the 436.1% cannot be re-derived later even by its author.

RECOMMENDATION

Add `transformers`, `torch` and the resolved model revision to the persisted JSON. Three lines, and it is the difference between a measurement and an anecdote.

4 The Igbo figure rests on five clips

BEFORE PUBLICATION

Igbo was run at 5 clips and 102 reference words; Hausa at 20 clips and 460. The two now sit side by side in the README and in the plan's progress table, formatted identically, with nothing in the presentation signalling that one has a quarter of the evidence behind it.

The Igbo run was a triumph — first successful fetch after weeks of failure — and the instinct to record it immediately was right. But 5 clips with 1 collapse gives a collapse rate of 20% with an interval wide enough to contain almost anything.

RECOMMENDATION

Re-run Igbo at 20 clips to match, or mark the sample size wherever the figure appears. The corpus is already local, so this is minutes.

5 The collapse threshold has no sensitivity check

BEFORE PUBLICATION

`Utterance.degenerate` defines a collapse as a hypothesis more than **3×** its reference length. The docstring is honest that this is a generous heuristic and explains the clip that motivated it. But the 35% collapse rate — now the project's headline result — is entirely a function of that constant.

```
scoring.py    return len(self.hypothesis) > 3 * max(1, len(self.reference))
```

A reviewer will ask what happens at 2× and at 4×. If the answer is “35% either way”, that is a strong result and cheap to state. If the rate swings, the threshold needs a defence rather than a comment.

RECOMMENDATION

Report the collapse rate at 2×, 3× and 4× once, from the existing JSON. No re-inference required — the hypotheses are already stored.

6 The plan PDF has no build recipe

HOUSEKEEPING

`docs/plan.pdf` is rendered from `docs/plan.html` with a headless-Chrome invocation recorded nowhere in the repository. The next regeneration is someone rediscovering the flags, and a PDF that drifts from its HTML source is worse than having only one of them.

RECOMMENDATION

A three-line `make plan` target, or a comment at the top of `plan.html`.

04 Coverage

What exists on disk against what has been measured.

LANGUAGE	ROLE IN THE DESIGN	CORPUS	EVALUATED
Hausa	Lead language; tone unwritten	734 MB · 621 rows	yes · 20 clips
Igbo	Tone-orthography arm	1,013 MB · 969 rows	partial · 5 clips
Yorùbá	Tone-orthography arm	827 MB · 831 rows	never run
English	Control arm	383 MB · 647 rows	never run

All four corpora are local and parse — 2.9 GB, fetched by `scripts/fetch-corpus.sh` with an append-only range loop against a pinned revision. The data half of Milestone 2 is genuinely unblocked; the measurement half is half-done.

05 What is done well

Stated because a review that lists only deficits misrepresents the work.

Pooled, not averaged

The corpus figure comes from one `jiwer` call over every pair. A per-utterance mean would weight a three-word clip like a thirty-word one, and short utterances are exactly where a silent model scores worst. The docstring says so.

Excluded figures never replace the full ones

`cer_excluding_degenerate` returns `None` when nothing was excluded, so the two can never be confused. The comment — “dropping inconvenient utterances silently is how a benchmark becomes an opinion” — is the right instinct, enforced in code rather than asserted in prose.

Comments record the failure, not the behaviour

Three of the hardest bugs are documented where they bit: the pipeline mutating its input dict (surfaced on clip 14 of 20), `return_timestamps` being a pipeline parameter rather than a `generate_kwargs` key, and the read-before-join ordering that stops a multi-megabyte payload deadlocking on the pipe buffer.

The fetch deadline is enforced from outside

A HuggingFace fetch can block below the level at which Python delivers signals — a `signal.alarm(90)` probe ran past 330 seconds. The child-process design is the correct response, and the docstring records the measurement that forced it.

Empty references are dropped, not scored

WER divides by reference length, so a blank reference is undefined rather than perfect. Handled explicitly, with an actionable error when *every* reference is blank pointing at a schema change.

06 Before Milestone 2

Ordered by lead time, longest first.

ITEM	WHY NOW	LEAD TIME
Recruit native-speaker reviewers	§8 calls this blocking for publication. Orthographic decisions must be reviewed before the normaliser is trusted — and if those calls are wrong, every normalised figure is wrong.	weeks
Choose the noise corpus	Defines what C3 means. §8 requires realistic market, street or babble noise, not white noise, with the source documented. Shapes the whole degradation pipeline.	a decision
Accept Common Voice terms	§8 flags it as likely gated. Five minutes now; a blocked milestone later.	minutes
Run English and Yorùbá	Finding 1. The baseline should predate the grid.	hours
Commit the results, record the stack	Findings 2 and 3. Seven conditions × four languages multiplies both problems.	an hour

Only the degradation pipeline itself is pure code, and it is the one piece with no external dependency — `librosa` and `soundfile` are already in the lockfile. It is also where the most interesting open question sits: **does the 35% collapse rate rise under narrowband and noise?** If it does, the failure mode found on clean audio becomes a considerably stronger result.